

WASP-80b

R-2277

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-80_210927 · created 2026-08-16T23:44:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459484.7201 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.011
Transit depth (Rp/Rs)^2	2.39 +/- 0.33 [%]
Orbital Inclination (inc)	88.8 +/- 0.77
Airmass coefficient 1 (a1)	0.978 +/- 0.011
Airmass coefficient 2 (a2)	0.0152 +/- 0.0048
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	#1 - [471, 127]
Optimal Aperture	4.67
Optimal Annulus	9.22
Transit Duration (day)	0.086 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.011 vs 0.1714 (deviation 0.99σ)
Duration fit vs published	0.086 d vs 0.0892 d (deviation 0.6σ)
Mid-time O–C	5.2 min
Depth SNR	9.4
Residual scatter	1.15 %

Light curve

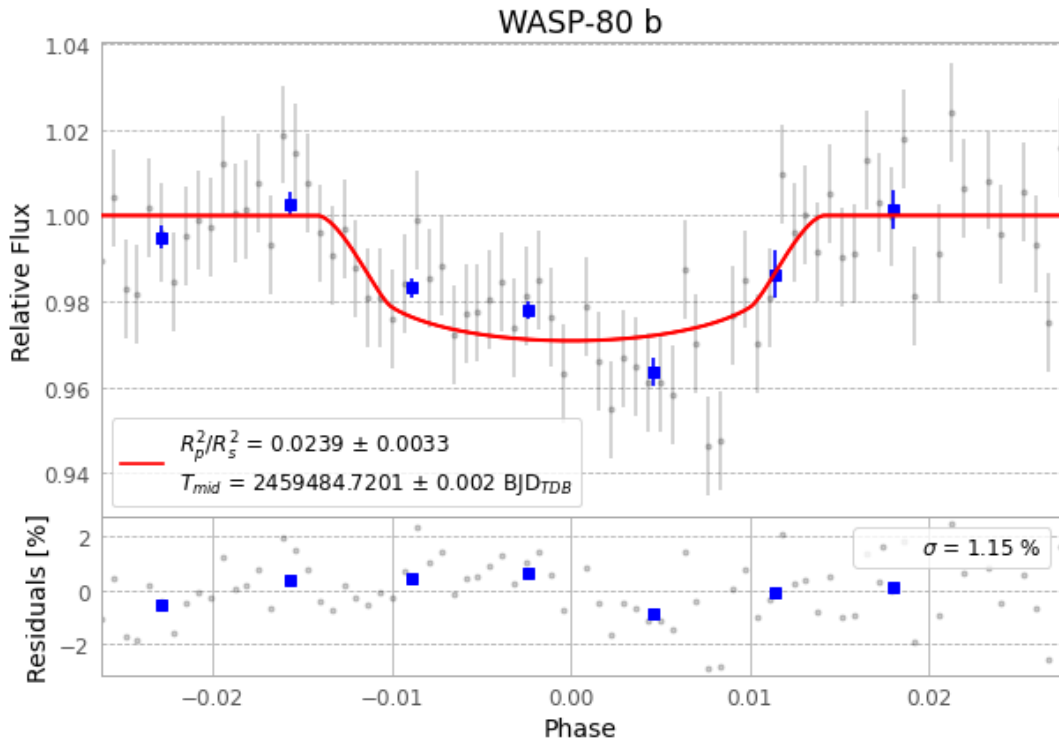


Figure 1 — FinalLightCurve_WASP-80 b_2021-09-26.png (SHA-256 933f5e47e60f0594...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 111], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 290eca99006ef480...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

80 FITS frames (53 MB), WASP-80210927031513.FITS → WASP-80210927071212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-210927.log (90ea2637247b...), FinalLightCurve_WASP-80 b_2021-09-26.png (933f5e47e60f...), FinalLightCurve_WASP-80 b_2021-09-26.csv (62f61069eb62...)

Compute resources

Wall time 118.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 23:44Z.